EXPERIMENT - 1 DATE : 20/07/26

AIM : STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS

⇒ WINDOWS COMMANDS

i) arp -a : Short form for address resolution protocol. It will show the IP Address of your computer.

OUTPUT :

(172.16.8.127) at 00:27:0e:13:f7:8c [ether] on enp3s0
(172.16.8.149) at 00:27:0e:13:f3:fe [ether] on enp3s0
(172.16.8.212) at 7c:5T:58:38:da:a3 [ether] on enp3s0
(172.16.8.186) at 00:27:0e:13:f5:77 [ether] on enp3s0
(172.16.11.207) at 7c:57:58:38:46:87 [ether] on enp3s0

ii) Hostname : Simplest of all TCP/IP commands. Displays name of computer

OUTPUT :
localhost.localdomain

iii) ~~nbtstat -a : Helps solve~~

iii) ipconfig /all : Displays detailed config. info about your TCP/IP connection

OUTPUT :

1 : lo : inet 127.0.0.1 scope host lo

2 : enp3s0 inet 172.16.8.127/22

iv) nbtstat -a : Helps solve problems with NetBIOS name resolution.

OUTPUT :

Resolved by Broadcast = 0

Resolved by Name Server = 0

Registered by Broadcast = 2

Registered by Name Server = 0

v) nslookup : A tool used to perform DNS lookups in Linux.

OUTPUT :

Server : 172.16.8.1

Address : 172.16.8.1#83

Non-authoritative answer:

Name : www.google.com

Address : 142.251.157.119

Name : www.google.com

Address : 142.251.156.119

vi) pathping : Combination of ping and Tracert command

OUTPUT : Over a max of 80 hops.

0 srujis [172.16.97.153]

1 srujis [172.16.97.153]

Computing statistics of 25 seconds

Hop	RTT	Source to here	This Node/Link.	Address
		host / sent = pct	host / sent = pct	
0				172.16.97.153
1	0ms	0/100 = 0%	0/100 = 0%	172.16.97.153

Trace Complete

vii) ping: Packet INternet groper commands is the best way to test connectivity b/w two nodes.

OUTPUT:

PING localhost (localhost (::1)) 56 data bytes
64 bytes from localhost (::1): icmp_seq=1 ttl=64 time=0.092 ms
64 bytes from localhost (::1): icmp_seq=2 ttl=64 time=0.043 ms
64 bytes from localhost (::1): icmp_seq=3 ttl=64 time=0.044 ms

⇒ LINUX COMMANDS:

~~i) ip: Most basic command used daily. Can show~~

i) ifconfig: Is a staple in many sysadmin's tool belt for configuring / troubleshooting networks

OUTPUT:

enp3s0: flags = 4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 1500
inet 172.16.8.127 netmask 255.255.252.0 broadcast 172.16.11.255
inet6 fe80::a6fb::cdb1::131c::6e50 prefixlen 64 scopid 0x20
ether 00:27:0e:13:f6:d9 txqueuelen 1000 (Ethernet)

ii) mtr: Serves as a network diagnostic tool

OUTPUT:

mtr google.com
local host: local domain ()

Host	Packets		Loss %	Snt	Last	Avg	Best	Wrst	StDev
				17.8	0.1	0.1	0.0	0%	0.0

iii) tcpdump - Designed to capture and display packets

OUTPUT :

tcpdump -D

1. enp3s0 (up, Running)
2. any [pseudo : device that captures all interfaces]
3. lo (up, Running, Loopback)
4. o/p 3 to 5
5. Bluetooth 0 (Bluetooth adapter number 0)

iv) pathping -4 172.16.97.153

OUTPUT :

over a

RESULT :-

Succefully executed Linux and Windows Commands